Take Home Assignment

Problem Statement

You will be given a dataset of conveyor belt images containing raw material particles. The dataset is unlabelled. You are expected to build a generalized segmentation model that can detect and delineate individual particles, along with a Python inference pipeline that processes an entire directory of images and returns the results.

Dataset

- Images are captured from overhead cameras mounted above industrial conveyor belts.
- The dataset contains no labels. You are free to generate annotations using any approach you see fit.

Inference Pipeline

Deliver a Python script (pipeline.py) with the following interface:

```
python pipeline.py --image_dir <path_to_image_folder> --output_dir <folder>
```

Expected Output

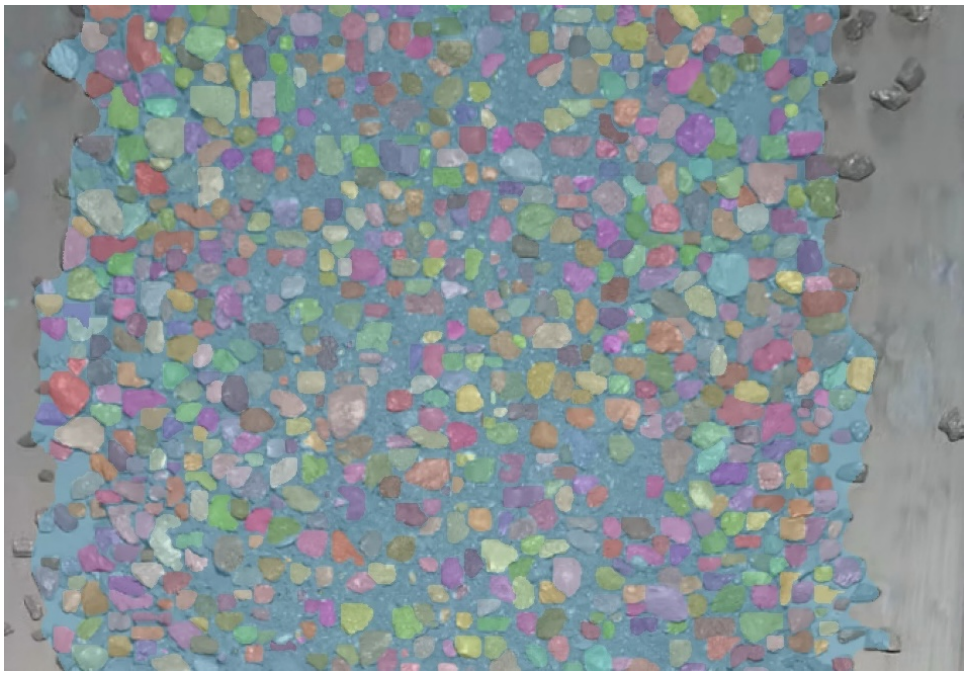
For each input image, the pipeline must produce the following:

Output	Description
`Annotated Image`	Original image with coloured instance masks overlaid on each detected particle, saved as <original_image_name>.jpg. See example below.

`detections.json`	One JSON file per image. Each key is a sequential particle index (order is arbitrary). Format:
	`{`
	`  "1": {`
	`  "bbox_coordinates": [x_min, y_min, x_max, y_max],`
	`  "mask_area": <pixel area of mask>`
	`  },`
	`  "2": { ... },`
	`}`
	bbox_coordinates: [x_min, y_min, x_max, y_max] in pixels. The annotated image is saved as <original_image_name>.jpg and the corresponding JSON as <original_image_name>.json. The particle index key holds no significance beyond identification.

Sample Annotated Image

The following is an example of the expected annotated image output:



Each detected particle is overlaid with a unique coloured instance mask.

Deliverables

Item	Description
`pipeline.py`	Inference script — clean, modular, runnable from CLI
`train.py / notebook`	Training script with instructions to reproduce
`Model weights`	Trained model weights (shareable link if large)
`Annotations`	Generated labels used for training
`outputs/`	Annotated images + detections.json for all dataset images
`README.md`	Setup, training, and inference reproduction steps